DOCUMENT CONTROL & METADATA			
Project Name:	Flow Analysis for Internal Pipes	**Lead Analyst:**	AVX Engineering Team
Simulation Type:	CFD Analysis	**Software Tool:**	SolidWorks 2026
Revision:	v1.0	**Date**	Sept. 06, 2026

1. Introduction

This technical report presents the Computational Fluid Dynamics (CFD) flow analysis conducted on the internal piping network supplied by the ZS 5 VSD 132 kW J-1.0-60 air source. The objective of the analysis is to evaluate the airflow distribution, velocity characteristics, and pressure behavior throughout the piping system and diffuser network under the specified operating condition of 100 m^3/min inlet volumetric flow rate.

The simulation was performed to assess the capability of the existing air supply system to distribute airflow effectively across the entire network. Results indicate that the available airflow from the source is insufficient to maintain the required flow conditions throughout all sections of the piping system. As the air progresses from the supply source toward the downstream branches and diffuser outlets, significant pressure losses and velocity reductions are observed due to flow resistance, pipe friction, geometric changes, and distribution effects within the network.

The CFD results further demonstrate the presence of low-velocity and stagnant airflow regions within several sections of the piping layout. These regions represent areas where the available flow energy has substantially diminished, resulting in poor airflow propagation and limited air delivery to the corresponding diffuser locations. Consequently, airflow distribution across the system becomes non-uniform, with some sections receiving significantly lower flow rates than others.

Analysis of the velocity contours, pressure distribution, and flow streamlines confirms that a considerable portion of the supplied air energy is dissipated before reaching the terminal diffusers. The identified dead energy zones indicate that the current operating condition of 100 m^3/min is inadequate to effectively supply air throughout the entire piping network. As a result, the system experiences reduced airflow performance and diminished diffuser effectiveness in downstream locations.

The findings presented in this report provide a technical basis for evaluating the current system performance and identifying potential improvements to enhance airflow distribution, reduce energy losses, and achieve more efficient utilization of the compressed air supply.

2. Simulation Setup

SolidWorks CFD	
Analysis Type	Internal
Unit System	SI (m-kg-s)
Fluid	Air
Total Cell Counts Mesh	1,555,701
Flow Type	Laminar and Turbulent
Gravity	On
No. of Iterations	658
Calculation Time	4.2 Hours
Boundary Conditions	
Inlet Volume Flow	100 m^3/min or 1.666 m^3/s (Source: Specs sheet of the machine)
Environment Pressure	101325 Pascal

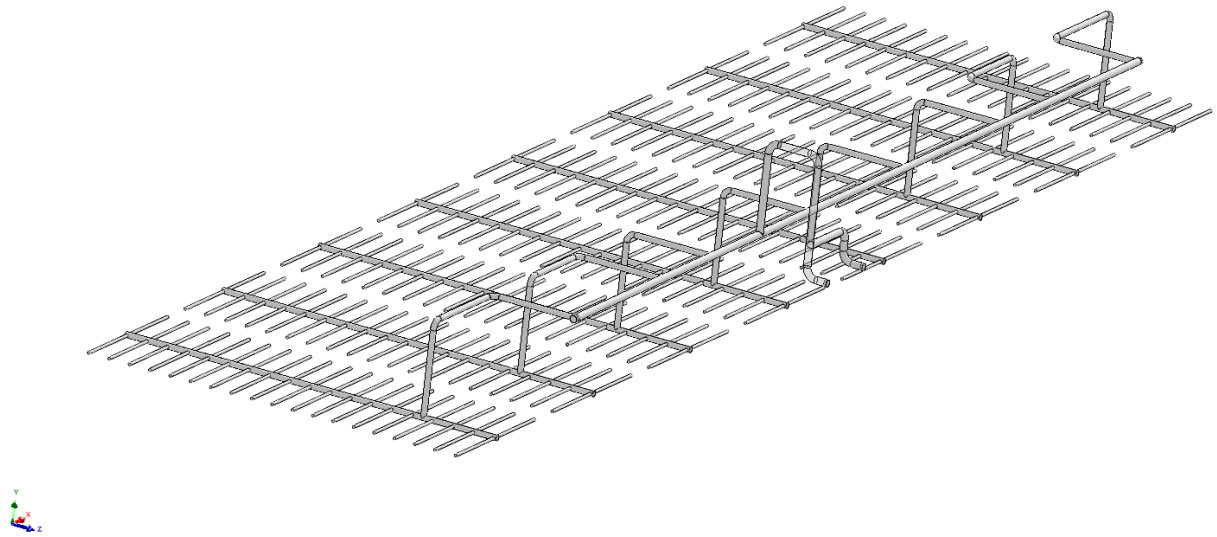


Figure 1. SolidWorks 3D Model

Presents the 3D solid geometry used to set up the CFD fluid domain. It defines all internal pipe surfaces, elbows, and diffuser outlets for computational meshing.

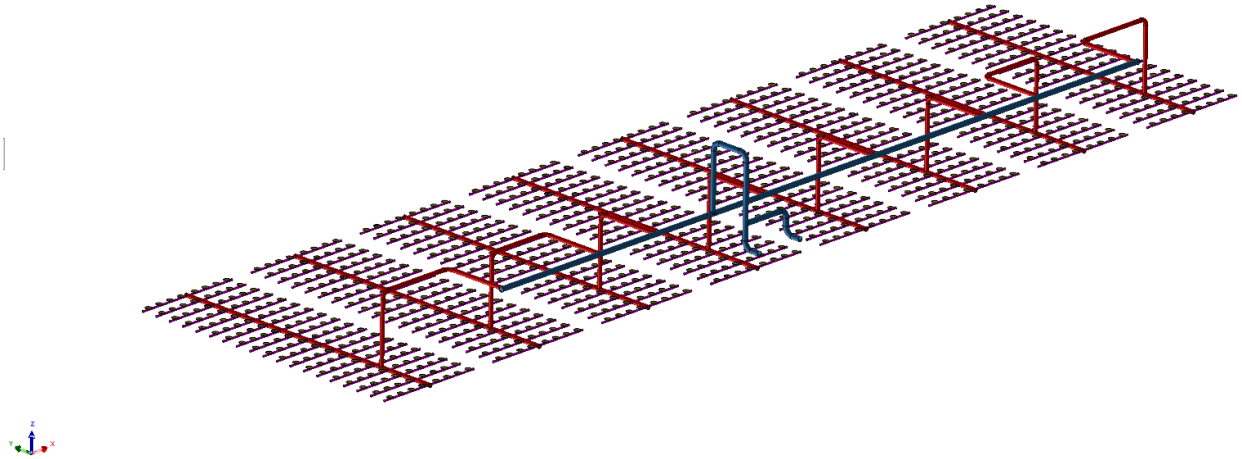


Figure 2. AutoCAD 3D Model

Shows the spatial orientation and layout dimensions of the piping system within the physical facility, verifying structural alignment and clearances.

3. Simulation Report

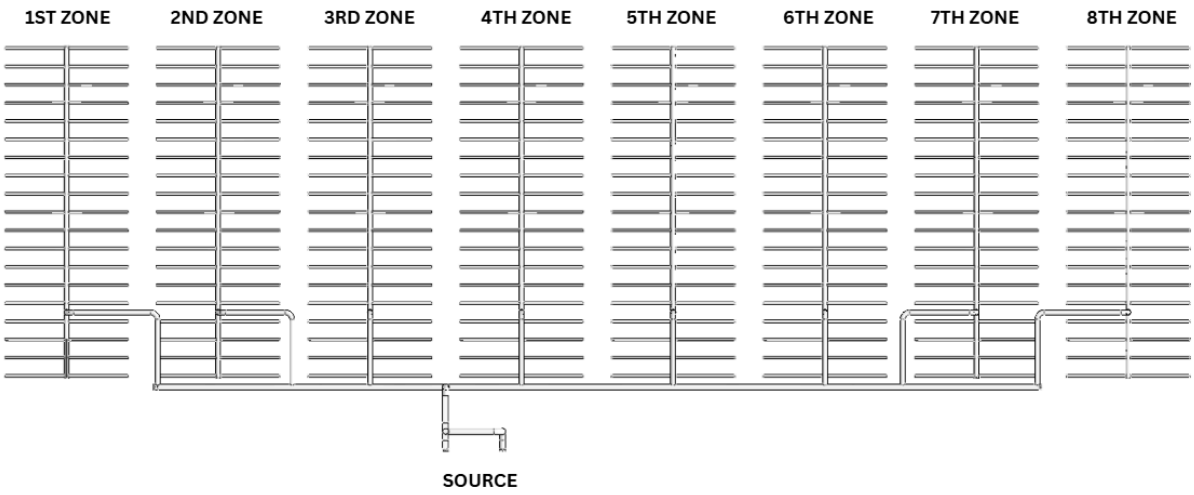


Figure 3. Zones Order

Maps out the network breakdown into distinct analytical sectors (Zones 1 through 8) to trace flow loss systematically from intake to end outlets.

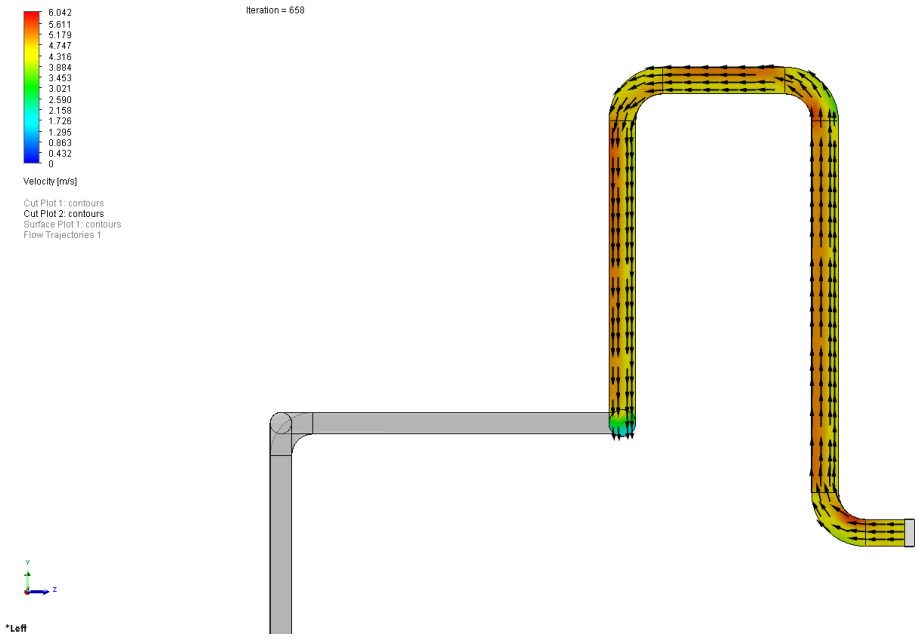


Figure 4. Air Inlet Pipe

Illustrates air entering from the primary supply source. A sharp elbow near the inlet causes immediate flow separation and turbulence, leading to an early loss in static pressure.

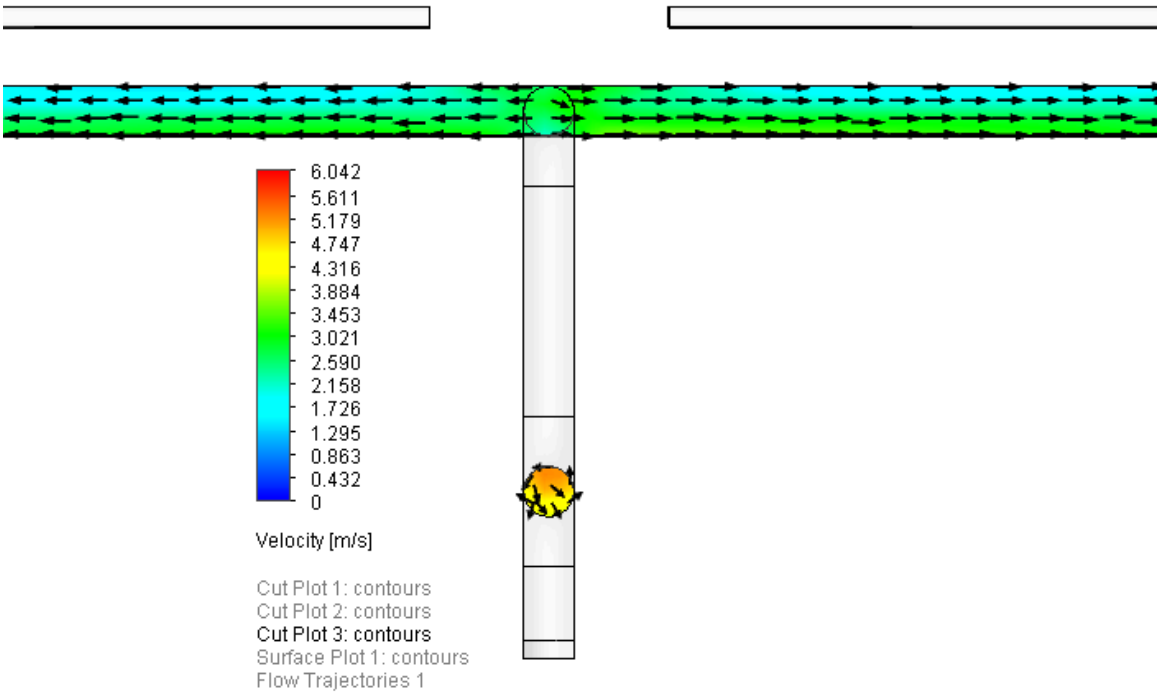


Figure 5. Top Split Line

Highlights flow behavior at the main junction where the single supply pipe splits into secondary branches. The sharp turn causes unequal air distribution across branches.

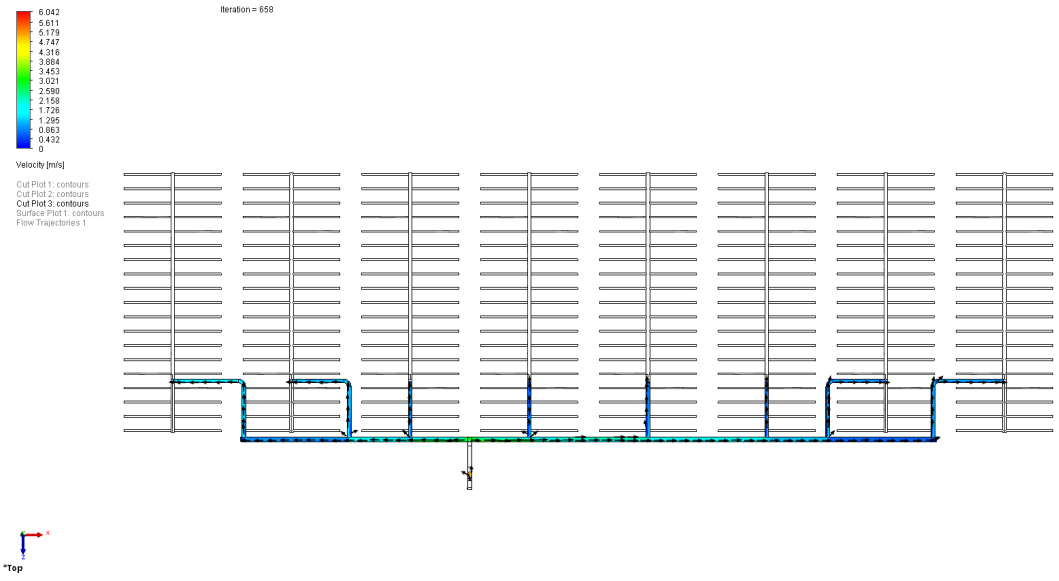


Figure 6. Top Pipes Cut Plot

Provides a horizontal cross-section showing air velocity and pressure levels across all main horizontal runs, showing gradual velocity drop over distance.

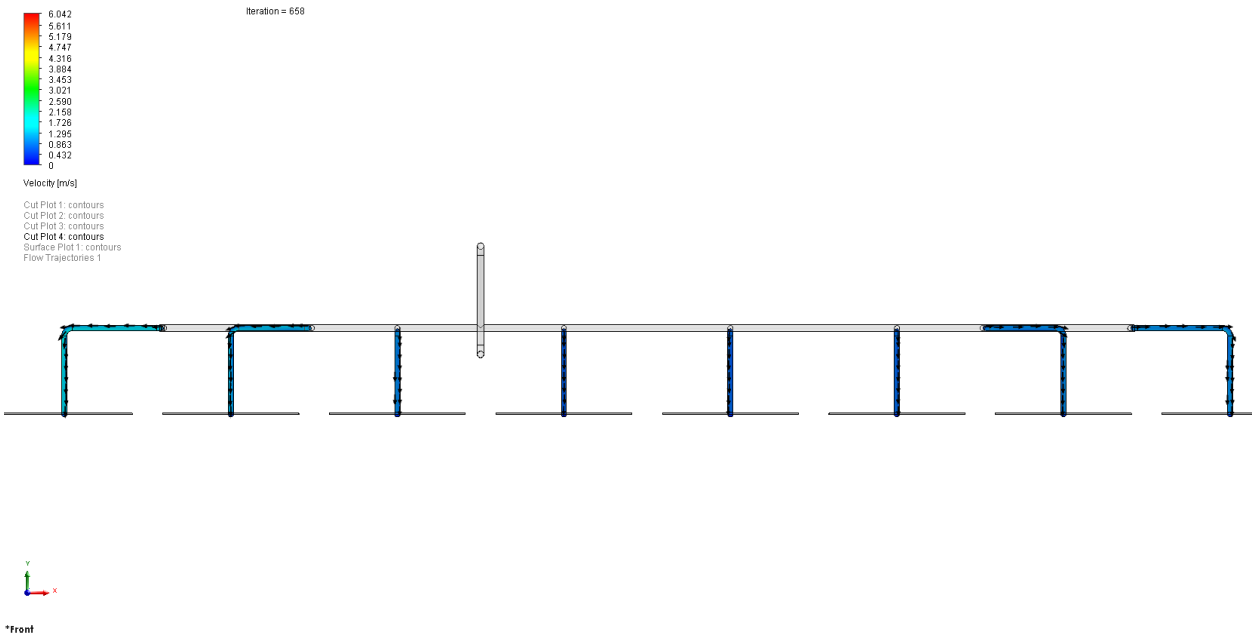


Figure 7. Front Pipes Cut Plot

Displays a vertical cross-section of downspouts and lower pipe sections, illustrating gravity and wall friction effects on vertical flow momentum.

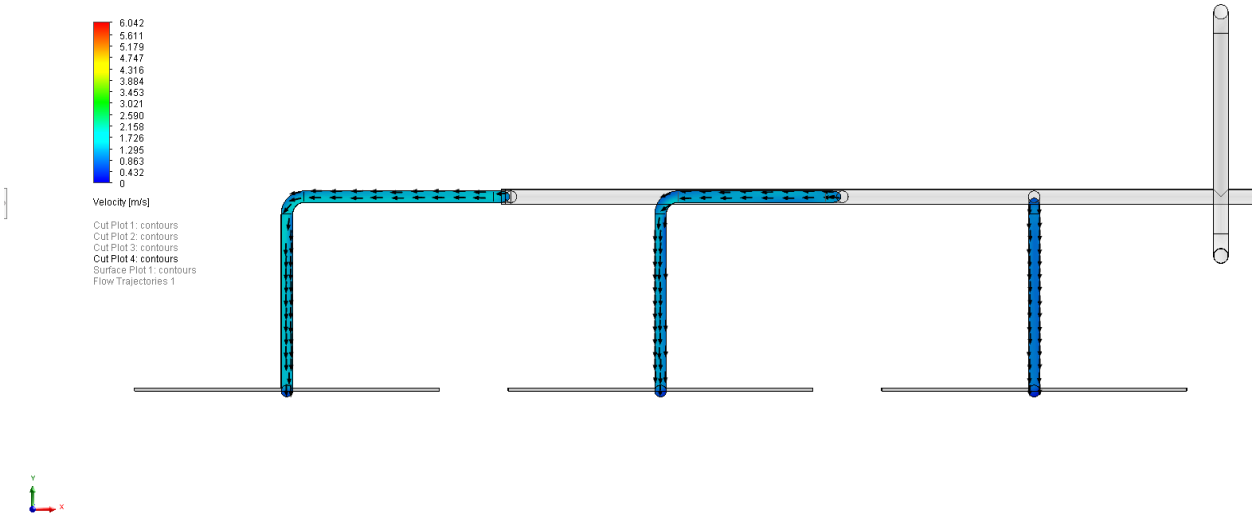


Figure 8. Left Potion Internal Pipes

Details internal flow profiles in the left wing. Repeated bends continuously reduce flow energy, decreasing local pressure before reaching diffusers.

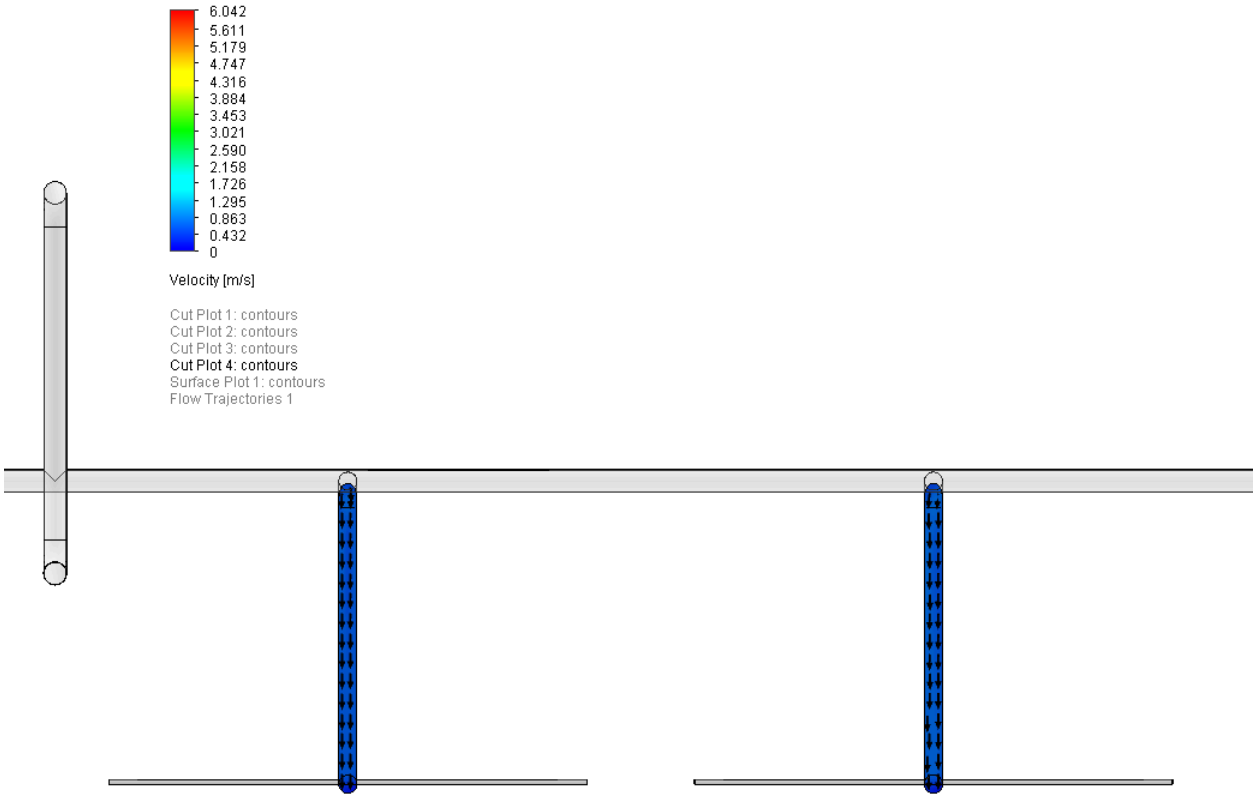


Figure 9. Inner Right Portion Cut Plot

Shows velocity profiles in the inner right pipeline, demonstrating a steady decrease in speed compared to the main header line.

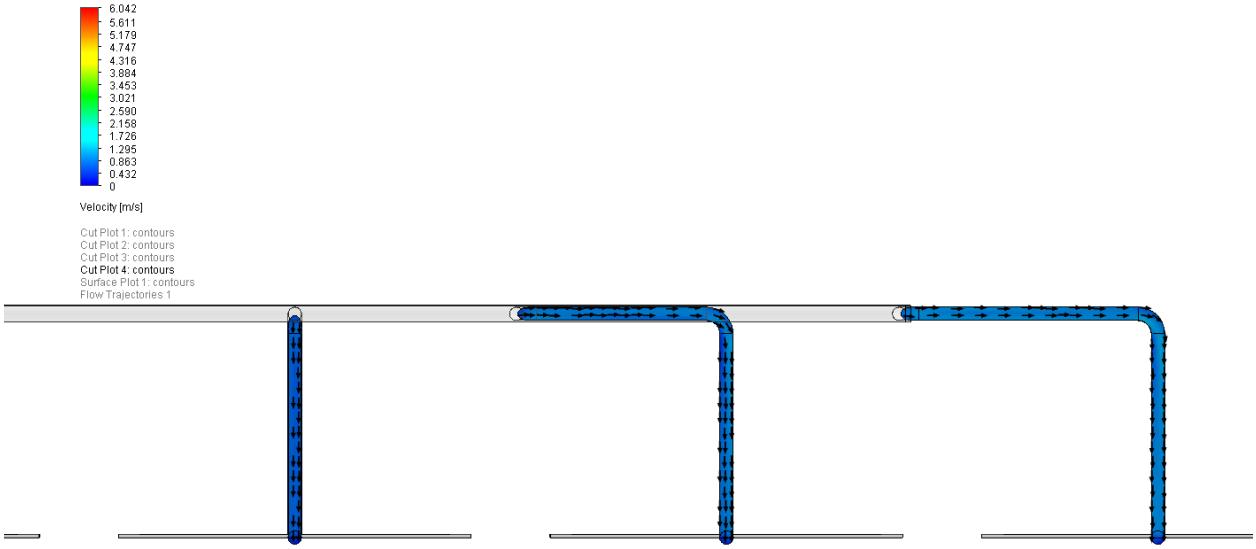


Figure 10. Outer Right Portion Cut Plot

Examines the outer right piping section, highlighting noticeable low-velocity zones caused by cumulative friction and distance from the air source.

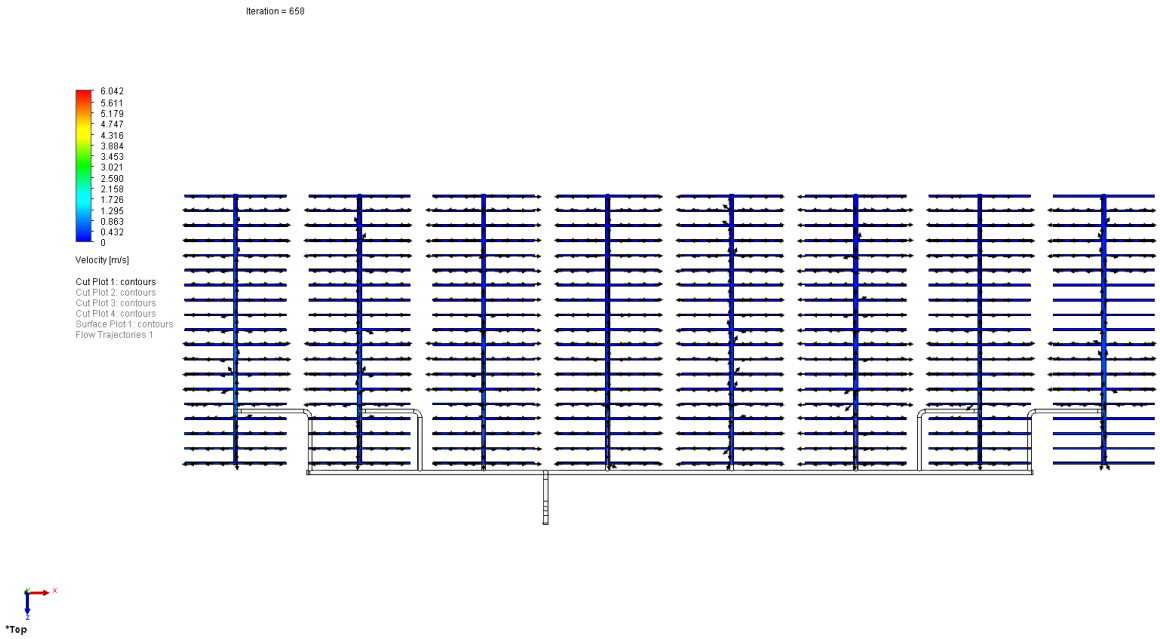


Figure 11. Bottom Diffuser Cut Plot

Analyzes air discharge at the bottom diffuser outlets, confirming uneven delivery rates where near-source diffusers outperform distant ones.

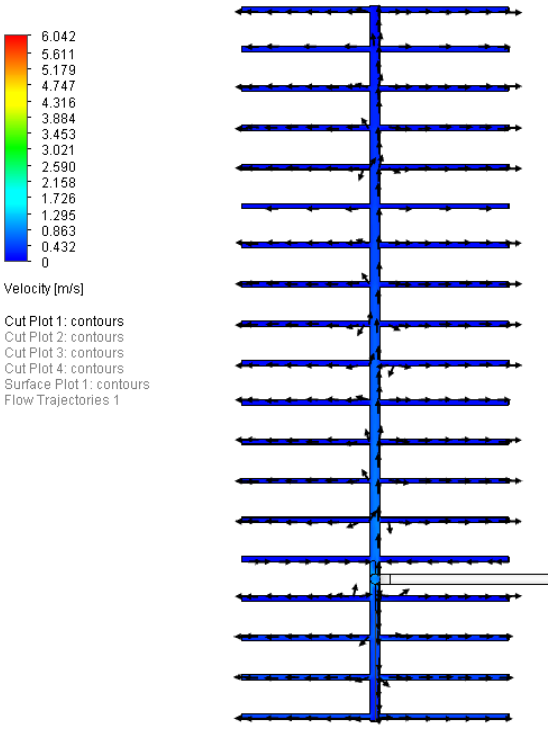


Figure 12. 1st Zone

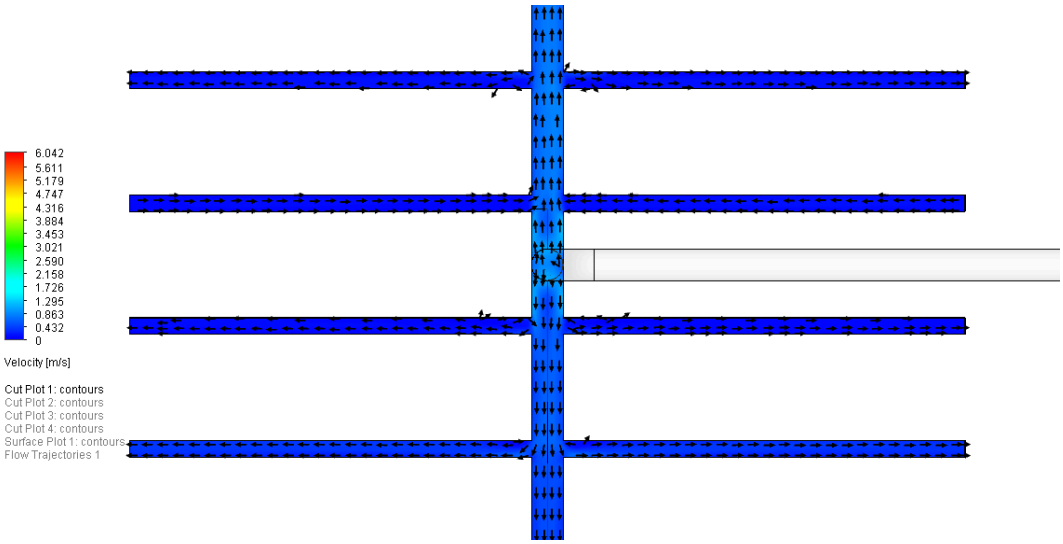


Figure 13. Detailed View - 1st Zone

Shows initial velocity air entering Zone 1. Minor turbulence forms around initial joints, but overall flow energy remains low.

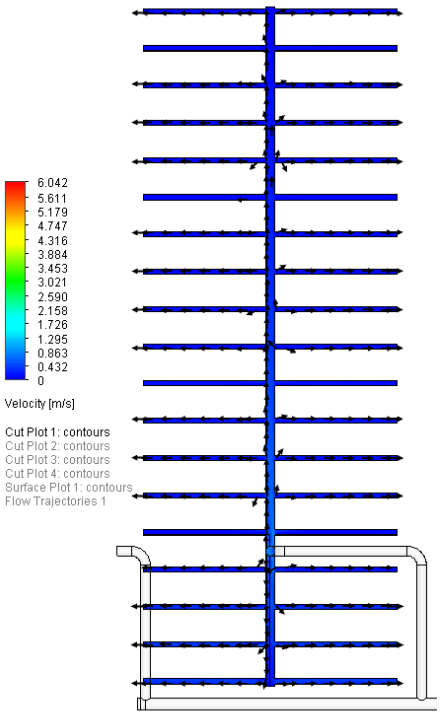


Figure 14. 2nd Zone

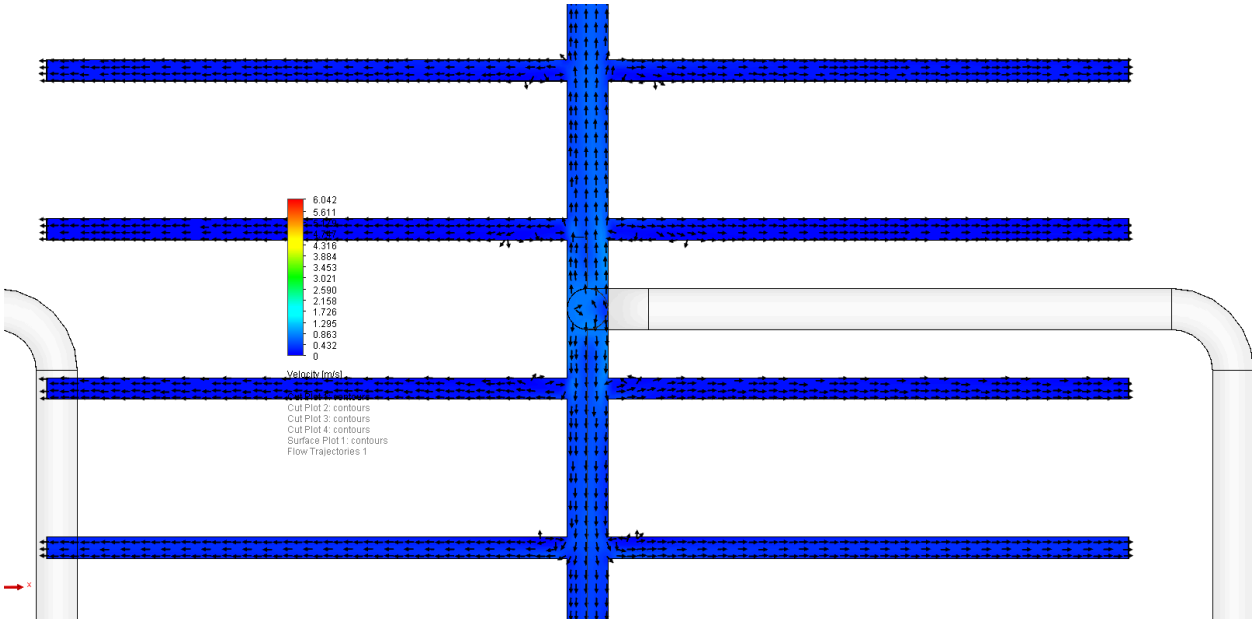


Figure 15. Detailed View - 2nd Zone

Captures air movement in Zone 2. As air moves through mid-stream bends, friction losses begin to reduce flow momentum.

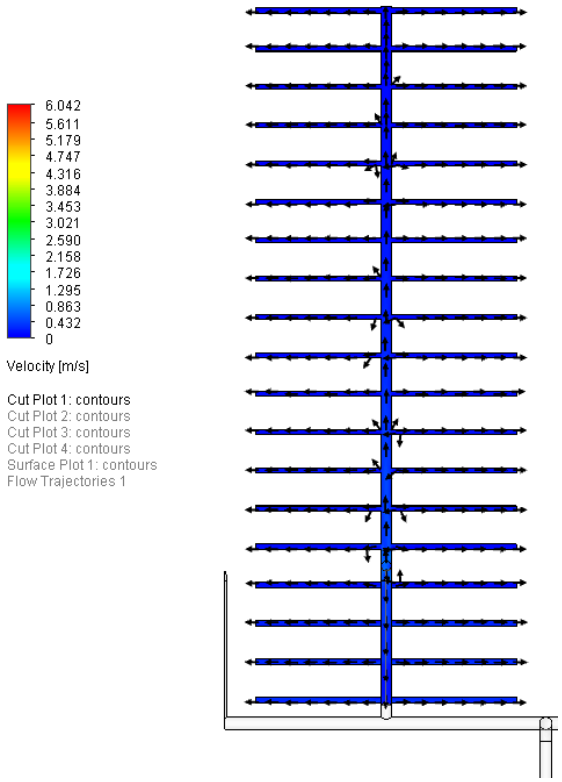


Figure 16. 5th Zone

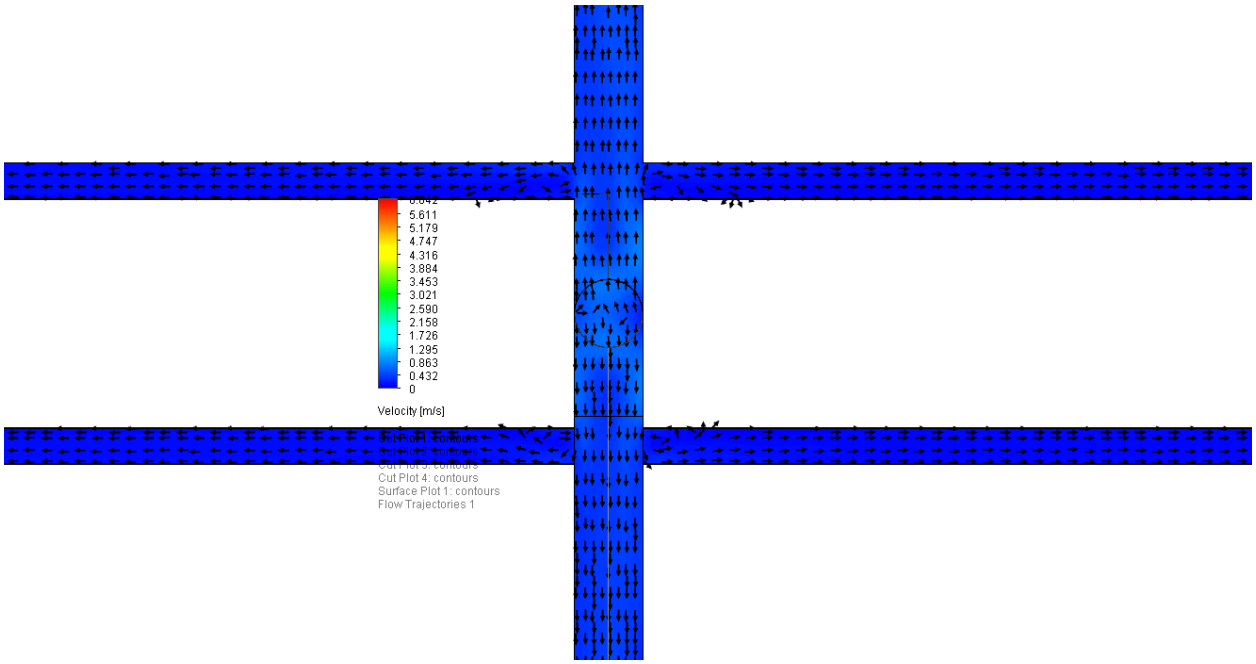


Figure 17. Detailed View - 5th Zone

Illustrates Zone 5 performance, where flow velocity drops significantly, creating early stagnant air pockets near outer pipe walls.

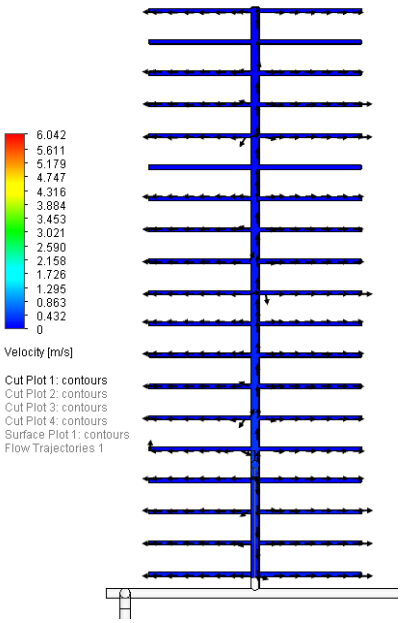


Figure 18. 6th Zone

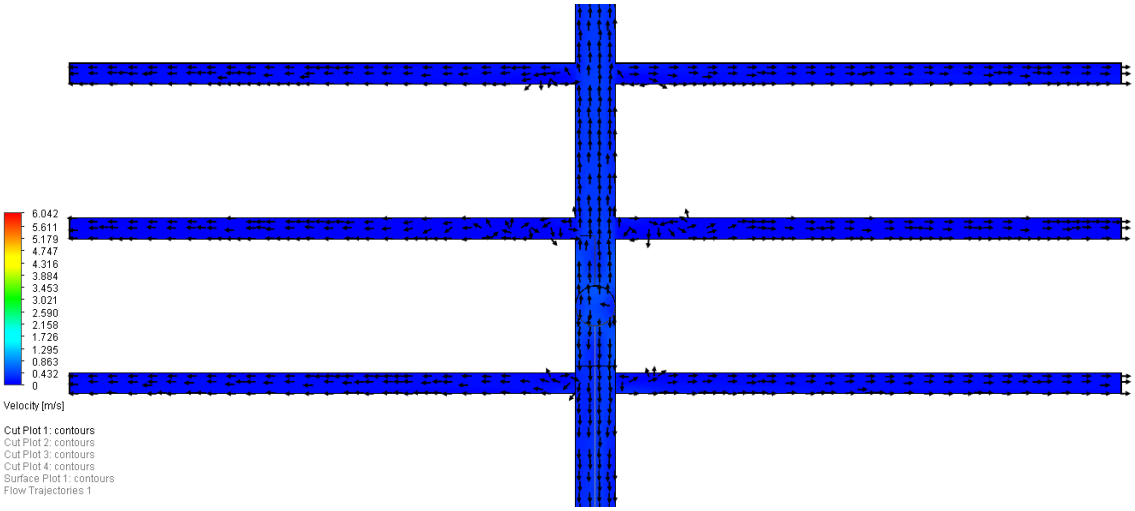


Figure 19. Detailed View - 6th Zone

Demonstrates severe pressure drops in Zone 6, with air velocity falling below effective operating speeds for downstream distribution.

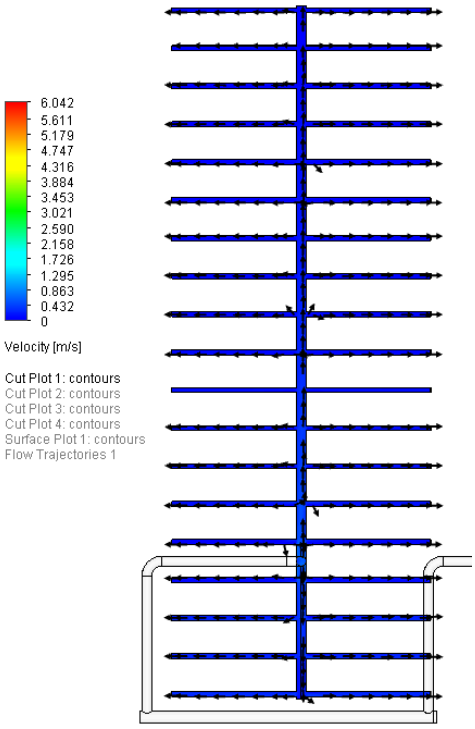


Figure 20. 7th Zone

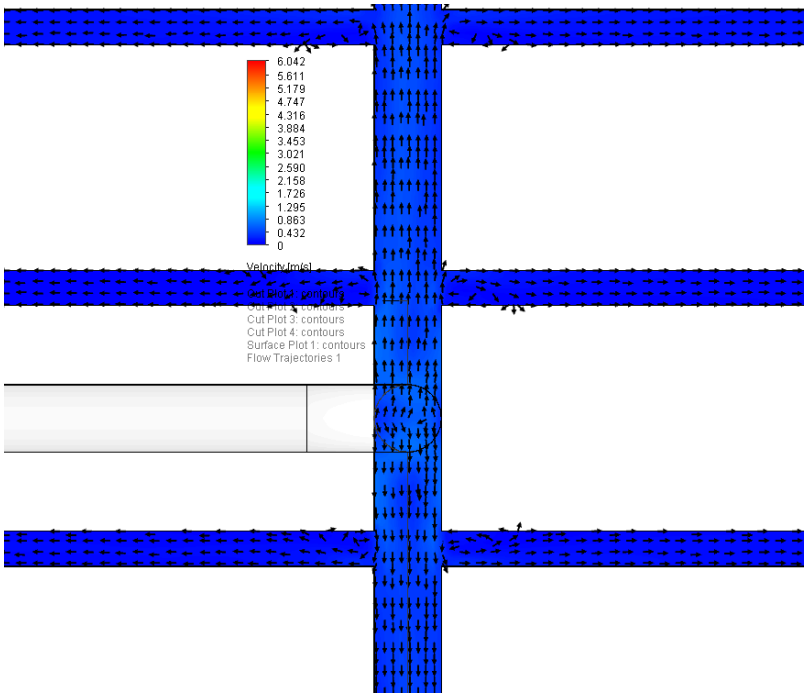


Figure 21. Detailed View - 7th Zone

Shows critical flow starvation in Zone 7. Low supply pressure results in weak discharge and localized air recirculation.

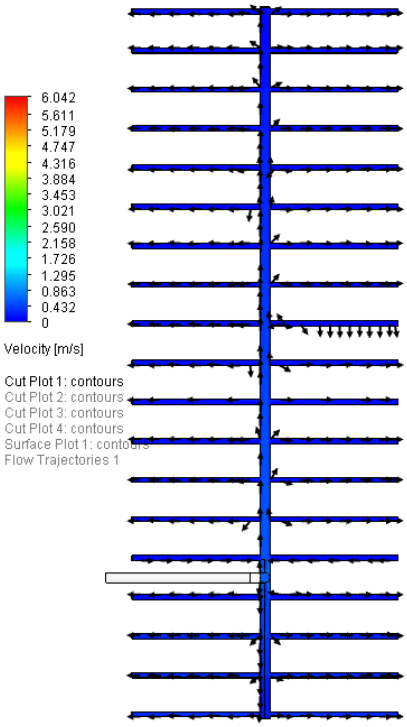


Figure 22. 8th Zone

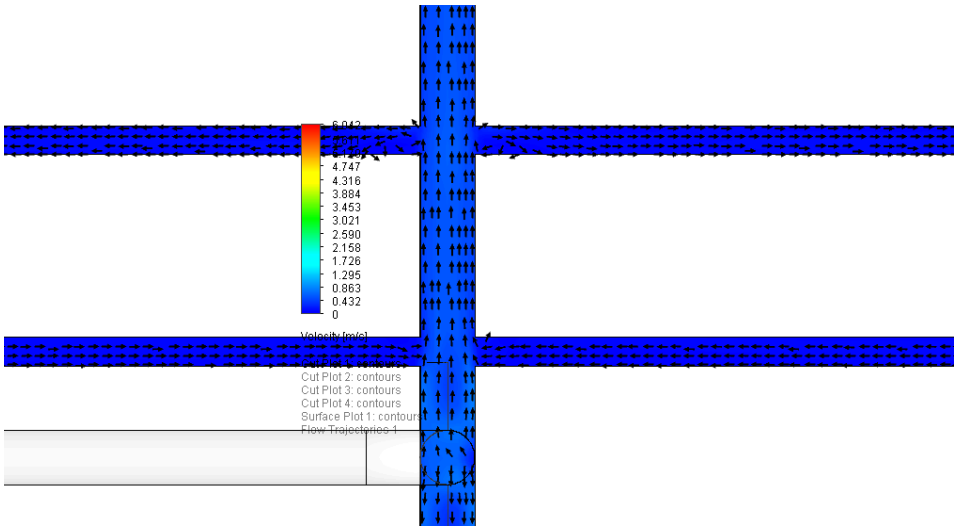


Figure 23. Detailed View - 8th Zone

Displays the farthest section of the network (Zone 8). Flow energy is entirely depleted, creating "dead zones" with zero effective diffuser output.

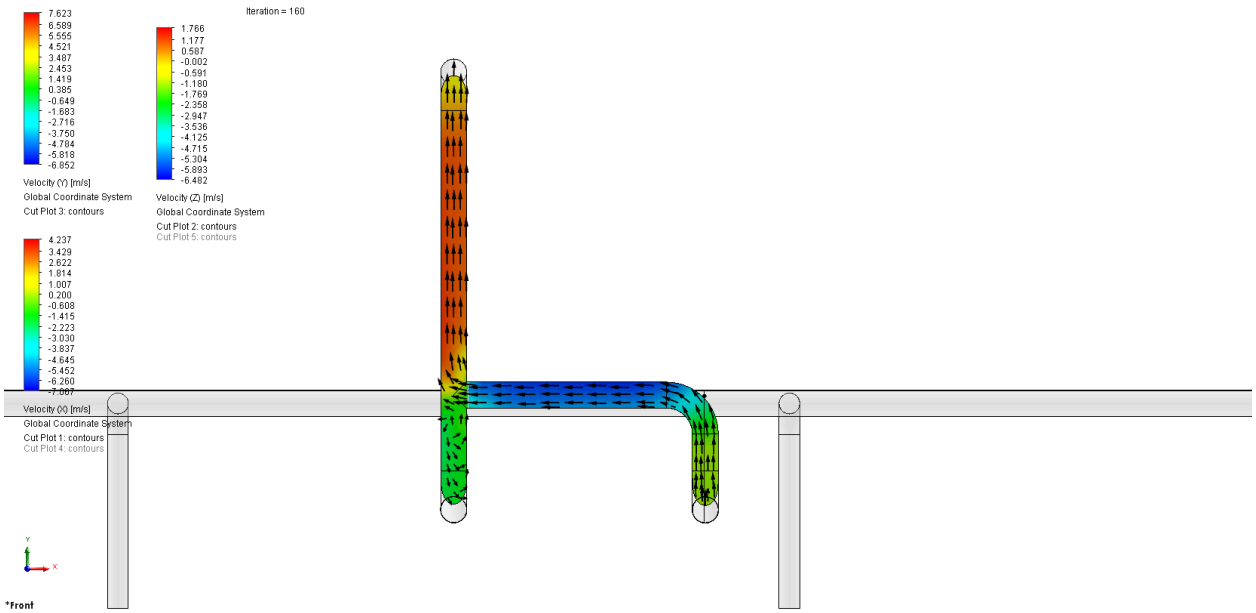


Figure 24. Additional Source Pipe

Shows the CAD modification working a second air supply line along the middle of the network to boost overall system pressure.

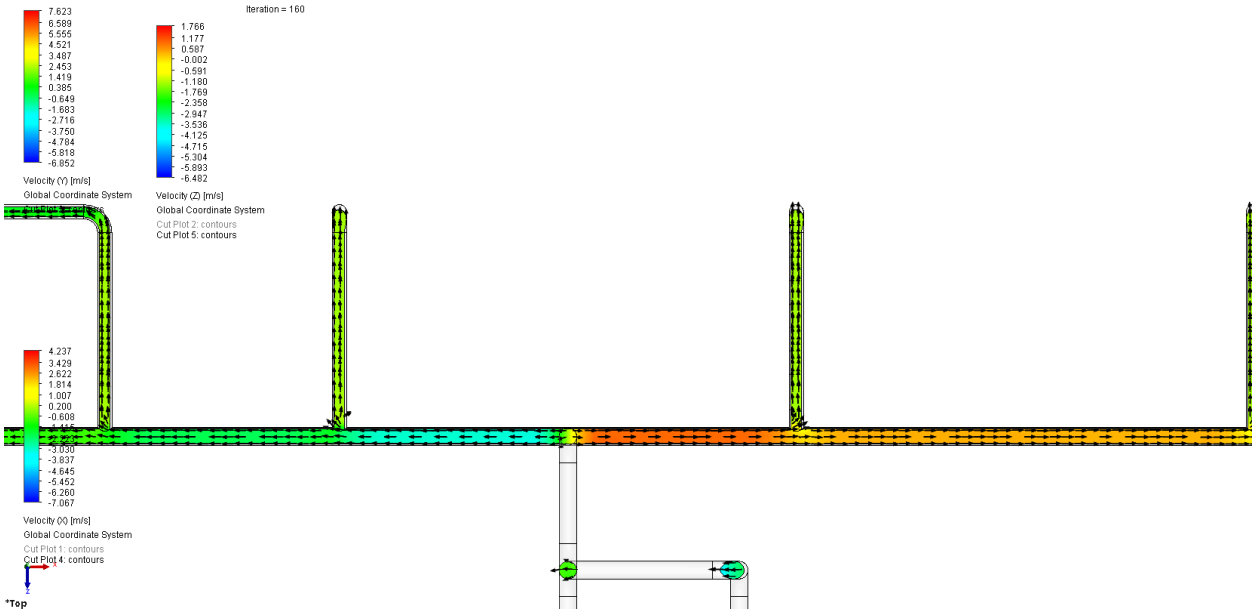


Figure 25. Top Cut Plot 2nd Source Pipe

Presents velocity cross-sections with the second pipe active, showing immediate pressure stabilization in central network branches.

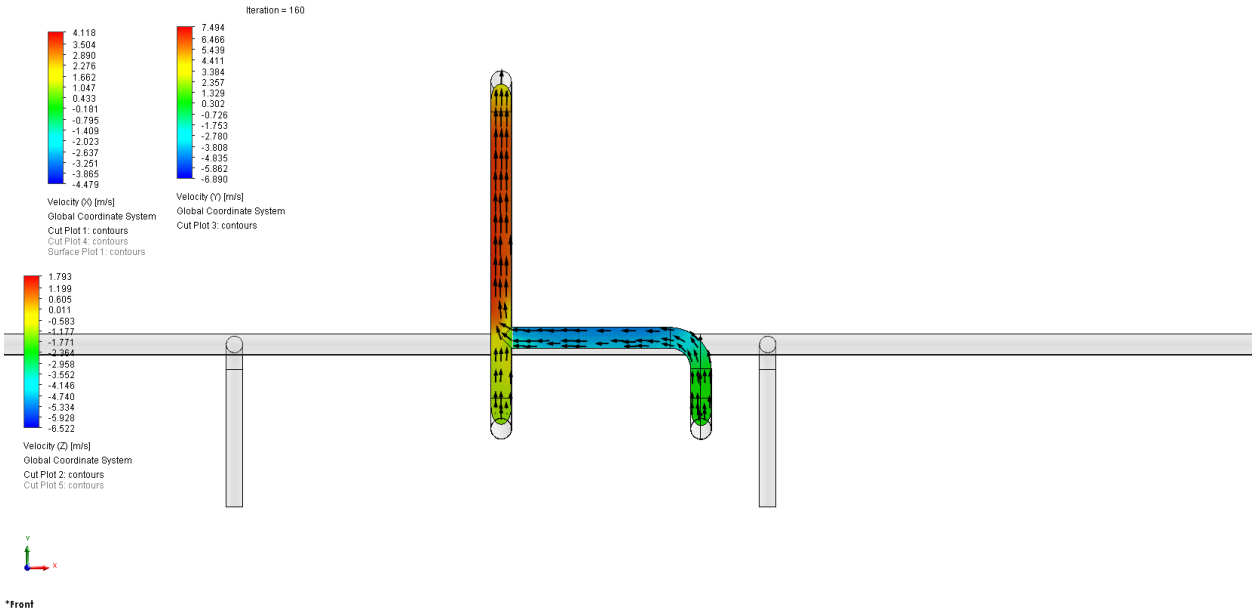


Figure 26. 2 Source Pipes Operating

Visualizes complete system streamlines under dual-source operation, demonstrating unified air delivery across all branches.

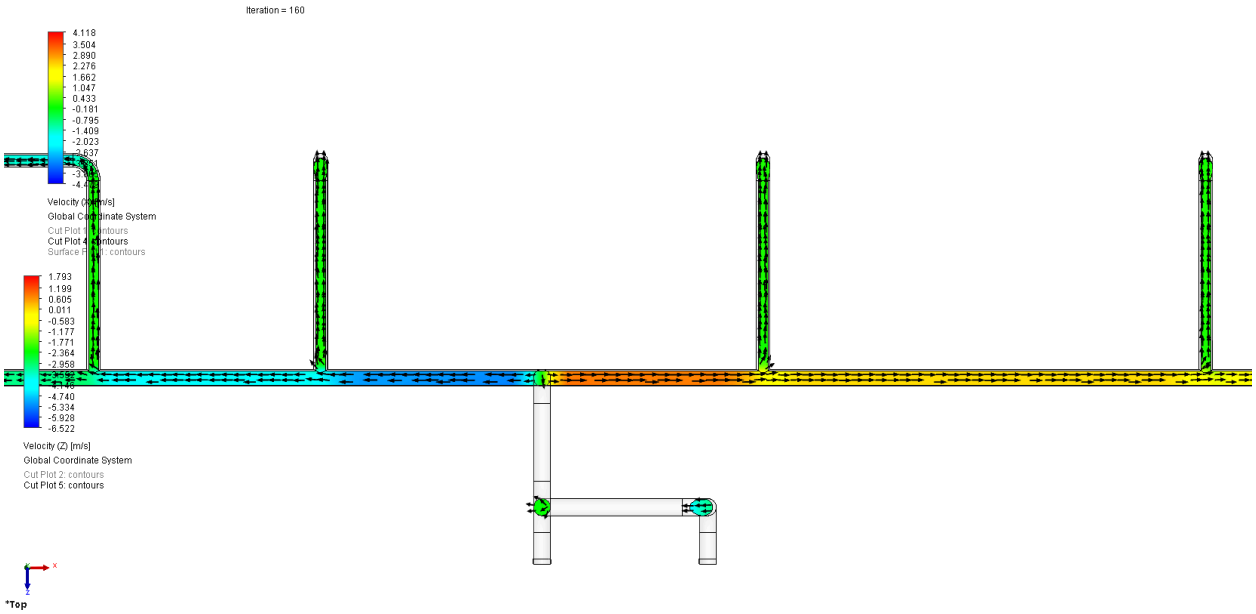


Figure 27. 2 Source Pipes Top Cut Plot

Shows high-level horizontal flow distribution under dual-source conditions, confirming the elimination of downstream dead energy zones.

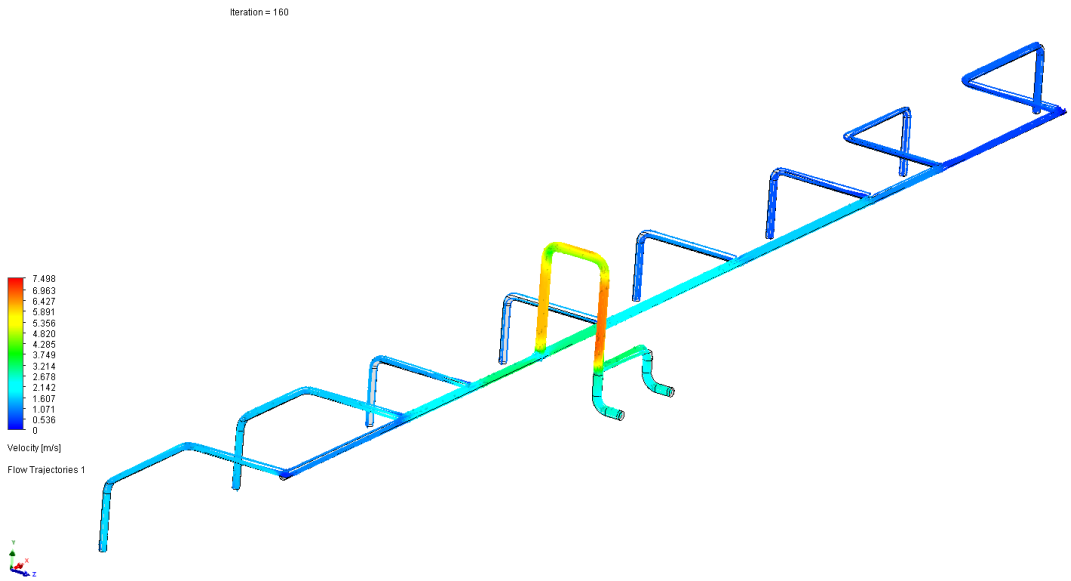


Figure 28. 2 Source Flow Graph

Graphically compares single-source vs. dual-source performance, proving a major increase in volume flow rate and pressure balance across all diffusers.

Video Source for Flow Animation: https://youtu.be/_bwmHDrHo1s

4. Analysis

Based on the simulation results, the current air distribution system is not able to efficiently deliver air to all diffuser locations. Airflow losses begin immediately after leaving the compressor due to the elbow installed near the air source. This bend disrupts the smooth movement of air, causing turbulence and reducing the air pressure available for the rest of the system.

As the air travels through the network, it passes through several bends and changes in direction. These features create additional resistance, causing the airflow to slow down and lose pressure before reaching the diffusers. As a result, the diffusers located farther from the source receive significantly less air than those located nearby.

The secondary air line connected from the middle machine also contains multiple elbows before joining the main pipeline. These bends further restrict airflow and may create unstable flow conditions, including localized backflow that can negatively affect system performance and place additional stress on connected equipment.

The simulation shows that a large portion of the air energy is lost within the piping network rather than being delivered to the intended diffuser outlets. Even if a higher-capacity air source is installed, the existing pipe arrangement may still prevent sufficient airflow from reaching the end sections of the system because the underlying restrictions remain unchanged.

If the current configuration continues to operate without modification, the system may experience increased vibration, noise, equipment stress, and premature wear of piping components. To improve performance, it is recommended to simplify the airflow path by reducing unnecessary bends and optimizing the pipe layout, allowing air to flow more efficiently throughout the entire system.

5. Recommendation

Based on the Computational Fluid Dynamics (CFD) simulation results, the existing air distribution network experiences significant pressure losses, air turbulence, and flow starvation in downstream sections. To ensure that all terminal diffusers across all 8 zones receive consistent, high-energy airflow, the following technical and mechanical recommendations have been formulated. These modifications focus on optimizing the piping layout, enhancing fluid dynamics, and implementing mechanical flow control mechanisms.

- **Primary Intake Bottleneck:** The initial elbow positioned immediately near the compressor outlet creates severe turbulence and a sharp static pressure drop right at the intake.
- **Cumulative Friction & Loss:** Long pipe runs combined with consecutive sharp bends continuously deplete air momentum, leaving downstream sections (Zones 7 and 8) starved of airflow and creating a "dead energy" zone.

A. Intake Pipe Geometry & Elevation Optimization

- **Extend Straight Inlet Length:** The initial straight pipe segment exiting the machine is currently too short. Extending this straight run before introducing any bends allows the air stream to stabilize into a smooth, fully developed flow, significantly reducing initial turbulence.
- **Eliminate Vertical Riser Pipes:** Remove the vertical raised pipe section located immediately after the inlet. Forcing air vertically upwards creates unnecessary backpressure and static losses due to gravity resistance.
- **Flow Recovery Space:** Because severe air turbulence disrupts flow velocity, longer straight pipe runs are required after turns to allow the air stream to settle, diffuse disturbances, and recover pressure.

B. Multi-Source Supply Architecture & Pipe Sizing

- **Zone Supply Ratio (2 Zones per Compressor):** Supplying all 8 zones from a single air source leads to severe energy depletion. It is recommended to reconfigure the system so that one compressor unit supplies approximately two (2) dedicated zones. This ensures sufficient volume flow and pressure across all diffuser outlets without dead energy zones.
- **Dual Intake Inlets per Zone:** Design each zone with two (2) separate air source inlet points to distribute airflow evenly across the local diffuser manifold.
- **Optimized Pipe Diameters:** Evaluate and increase pipe diameters along primary supply mains. Larger diameters decrease flow velocity relative to pipe walls, drastically reducing internal friction losses over long distances.

C. Bend Optimization & Fitting Geometry

- **Replace 90° Elbows with Swept/Long-Radius Bends:** Where facility boundaries require pipe turns, eliminate sharp 90-degree elbows. Replace them with long-radius curves or smooth angled bends (such as 45° sweeps) to minimize flow disruption and pressure drop.

D. Mechanical Control & Booster Enhancements

- **Integration of Mechanical Boosters & Governors:** Evaluate the addition of inline booster pumps/blowers and pressure governors at individual station zones. These mechanical components actively regulate supply pressure, recover pressure losses from upstream bends, and guarantee steady air delivery to terminal diffusers.

To successfully transition these recommendations from simulation findings into fully operational, high-efficiency physical infrastructure, **AVX Engineering Solutions** offers comprehensive engineering consultation and design service, AVX is open for service if you will consider:

Service Package	Scope of Deliverables
On-Site Assessment & Verification	Conduct a detailed technical site visit to inspect current physical spatial constraints, verify machine layouts, and evaluate real-world mounting points (ideal if the project is still in the planning or design phase).
Complete Engineering Design & CAD Drafting	Development of optimized 3D CAD piping models incorporating swept bends, balanced pipe sizing, optimized inlet elevations, and dual-source header connections.
Full Engineering Drawing Package	Provision of comprehensive 2D manufacturing drawings, system isometric plans, bill of materials (BOM), and mechanical installation specifications ready for fabrication and contractor execution.

Whether your team requires ongoing expert consulting or a complete turnkey engineering design package, AVX Engineering Solutions is ready to collaborate with your organization to achieve peak system performance.

NOTHING FOLLOWS